PA2-2026

CS300 - CSC13002 - Introduction to Software Engineering

Project Assignment 2 (PA2-2026)

Total scores: 60 points
Duration: 2-3 weeks
This is a teamwork assignment.

A - Project Plan (25 points)

You are required to draft an initial version of your project plan. After submitting this version, the TAs will review and provide feedback. Based on the feedback, you will update and submit again in the next assignment (PA3-2026).

The project plan should cover the following sections:

1. **Introduction:** brief overview of the project.
2. **Project Overview:** goals, scope, deliverables, and assumptions.
3. **Project Organization:**
    - Team structure and roles of each member.
    - Risk management: identify at least 3 potential risks (e.g., member unavailability, technology issues, scope creep) and describe mitigation strategies for each.
4. **Project Plan:**
    - This project applies the **Scrum** process. If you encounter references to RUP phases (e.g., Inception, Elaboration, Construction) in any provided templates, please disregard them.
    - The instructors recommend organizing the project into **5 Sprints**, each corresponding to one PA (Sprint 1 = PA1, Sprint 2 = PA2, etc.) and lasting 2-3 weeks. For the upcoming sprint, the team must provide detailed tasks with assigned members and due dates. For later sprints,

listing the planned tasks is sufficient; detailed assignments can be made when each sprint begins.

- Schedule: the schedule should be consistent with your Jira board.
- Build plan: specify how many builds you will produce for testing and when. Note that the final sprint (PA5) includes a demo of all project features, so plan your builds accordingly across the 5 sprints.
- All project-related activities — including report writing, coding, and self-training — are considered valid tasks. Each task should be performed by one member and reviewed by another member.
- Note that tasks within each sprint are not limited to the corresponding PA deliverables. Teams may include tasks for future PAs or additional coding tasks to ensure the project is completed on schedule by the end of the semester.

In this PA2-2026, the project plan is NOT required to be fully complete. Your team should provide all relevant information available at this stage. The plan will be revised in PA3-2026.

Write the project plan in Markdown format.

B - Vision Document (25 points)

Prepare an initial version of the vision document for your project. Similar to the Project Plan, this is an initial draft — you will revise the vision document in PA3-2026 based on TA feedback. The vision document should cover the following sections:

1. **Introduction:** purpose of the document and references.
2. **Positioning:**
   - **Problem Statement:** describe the problem your project solves (use the table format: the problem of / affects / the impact of which is / a successful solution would be).
   - **Product Position Statement:** describe your product's unique position (use the table format: For / Who / The product name / That / Unlike / Our product).
3. **Stakeholder and User Descriptions:**
   - Stakeholder summary, user summary, user environment.

- Summary of key stakeholder or user needs.
- Alternatives and competition (reference your existing app survey from PA1-2026; if your understanding of the project has evolved, update the PA1 survey accordingly).

4. **Product Overview:** product perspective, assumptions and dependencies.
5. **Product Features:** list and describe the product features. This section is a more detailed and refined version of the Key Features from your PA1 Project Proposal — features may be updated or adjusted based on what you have learned since PA1. For each feature, write a paragraph (3-5 sentences) describing what the feature does, why it is needed, and who benefits from it. Do not merely list feature names. Additionally, draw 1-2 workflow diagrams (using Mermaid format) that illustrate the most important user workflows of your application.
6. **Non-Functional Requirements:** list applicable standards, performance requirements, security requirements, platform requirements, and other quality attributes. Requirements should be quantifiable where possible (e.g., "page load time < 2 seconds", "support 100 concurrent users") rather than vague statements (e.g., "the app should be fast").

Write the vision document in Markdown format.

C - Spec Kit Initialization (5 points)

Set up a GitHub repository with Spec Kit integrated and initialize Spec Kit for your project by running the setup commands. The Spec Kit artifacts should be placed inside the `/src` folder of your repository. Submit the following artifacts generated during initialization:

- The `constitution.md` file created for your project.
- Any other Markdown files generated by Spec Kit during the initialization process.

Additionally, each team member must complete the Spec Kit training by following the official YouTube tutorials (links will be provided separately). Submit evidence of training completion (e.g., screenshots, training log, or a brief summary of what each member learned).

D - AI Usage Report and Weekly Report (5 points)

AI Usage Report

Each team must follow the AI usage guidelines to declare whether AI tools were used during this sprint. If AI tools were used, a detailed log of all usages must be submitted.

Weekly Report

Conduct and document the following meetings: Sprint Planning, Scrum meetings, and Sprint Review. Follow the instructions in the "WeeklyReports" document (provided separately).

Include Jira task screenshots showing task assignments and progress for this sprint.

Submission Guidelines

- All documents must be written in **English** using **Markdown** format. Diagrams should be drawn using **Mermaid** syntax.
- In each report/document section, the team must indicate who performed the work, who reviewed it, and who edited the section. Place this information on the first line immediately after the section header.
- Submit both the **original Markdown files (.md)** and **PDF files converted from Markdown**.
- Provide a **Git log** (screenshot or exported log) showing commit history of your repository.
- Compress all files into a directory named `PA2-Group[GroupId]` and submit as a zip file.
  Example: `PA2-Group01.zip`

Grading Criteria

The assignment will be graded according to how well your team prepares the required documents using the information your team has at this time. Criteria include:

- English writing: clear and easy to understand with few errors.
- Well-formatted texts, diagrams, and charts.

- Level of completeness of the documents, based on the information available at this time.
- The vision document must include a clear problem statement, product features with paragraph descriptions, and representative workflow diagrams.
- The project plan must include a realistic schedule consistent with Jira tasks.
- Spec Kit must be properly initialized with the required files submitted.